



Environmental Data Imputation via Temporal VAE with Learned Missing Value Representations

Vipin Kataria¹ , Nitin Kumar² , and Parth Patel³ 

¹ Picarro Inc, Santa Clara, CA, USA
vipink2@illinois.edu

² Marriott International, Dallas, TX, USA

³ Alindus Inc, Irving, TX, USA

Abstract. Environmental monitoring systems face persistent challenges with missing data, particularly in air quality networks where sensor failures, power outages, and maintenance activities frequently create gaps in temporal measurements. Traditional imputation methods struggle with the sophisticated temporal dynamics and complex relationships inherent in environmental data. This paper introduces a novel Temporal Variational Autoencoder (VAE) approach that addresses the fundamental limitation of standard VAEs in handling missing data through learnable missing embeddings. Instead of using arbitrary placeholder values for missing data, our method learns optimal representations for missing values during training, enabling VAEs to process partial observations naturally. We implement variable-type-specific learnable embeddings that account for different characteristics of meteorological versus pollutant variables, combined with a temporal VAE architecture optimized for environmental time series with multiple temporal scales. Comprehensive evaluation on EPA air quality datasets demonstrates substantial improvements across various missing scenarios, with 15–20% RMSE reductions compared to standard VAE approaches and 25–35% improvements over traditional statistical methods. The proposed method maintains computational efficiency suitable for deployment in operational environmental monitoring networks while providing uncertainty quantification crucial for downstream analysis and decision-making.

1 Introduction

Environmental monitoring systems face persistent challenges with missing data, particularly in air quality networks where continuous operation is critical for public health protection and environmental policy formulation. Sensor failures, power outages, extreme weather events, and routine maintenance activities frequently create gaps in temporal measurements, compromising the reliability of subsequent analyses and forecasting models [1]. The complexity of environmental data, characterized by strong temporal dependencies, seasonal patterns, and intricate relationships between meteorological and pollutant variables, makes effective imputation particularly challenging.

Traditional statistical imputation methods such as linear interpolation, mean substitution, and k-nearest neighbors often fail to capture the sophisticated temporal dynamics inherent in environmental systems [2]. While these approaches may suffice for short-term gaps under stable conditions, they struggle with longer missing periods and complex missing patterns that commonly occur in real-world monitoring networks. The emergence of deep learning techniques has opened new possibilities for more sophisticated imputation strategies that can better model the underlying temporal processes governing air quality dynamics.

Recent advances in deep generative models, particularly Variational Autoencoders (VAEs), have demonstrated significant potential for handling missing data through their ability to learn complex latent representations and generate realistic samples [3]. VAEs offer several advantages for time series imputation: they can capture non-linear temporal dependencies, provide uncertainty estimates through their probabilistic framework, and learn meaningful latent representations of temporal patterns. However, a fundamental limitation of standard VAE architectures is their assumption of complete input vectors, creating challenges when dealing with partially observed data during both training and inference.

Environmental data presents unique characteristics that make imputation particularly challenging. Air quality measurements exhibit multiple temporal scales including diurnal cycles driven by emission patterns and atmospheric chemistry, weekly patterns reflecting human activity, and seasonal variations influenced by meteorological conditions and photochemical processes [4]. Additionally, the data contains complex relationships between variables, such as temperature-dependent ozone formation, wind-driven pollutant dispersion, and humidity effects on particulate matter dynamics. These characteristics demand imputation methods that can preserve temporal structure while handling partial observations naturally.

This paper introduces a novel approach that addresses the fundamental limitation of VAEs in handling missing data through learnable missing embeddings. Instead of using arbitrary placeholder values (such as zeros or means) for missing data, our method learns optimal representations for missing values during training, enabling VAEs to process partial observations naturally. Our key contributions include: (1) A learnable missing embedding strategy that enables VAEs to handle partial observations during both training and inference; (2) Variable-type-specific embeddings that account for different characteristics of meteorological versus pollutant variables; (3) A temporal VAE architecture optimized for environmental time series with multiple temporal scales; (4) Comprehensive evaluation demonstrating significant improvements over existing methods across various missing scenarios and temporal patterns.

The proposed method offers both theoretical innovation and practical applicability. By solving the partial observation problem in VAEs, we enable effective temporal pattern learning even with substantial missing data. The approach maintains computational efficiency suitable for deployment in operational environmental monitoring networks while providing uncertainty quantification crucial for downstream analysis and decision-making.

2 Related Work

2.1 Variational Autoencoders for Missing Data Imputation

Variational Autoencoders have emerged as powerful tools for missing data imputation due to their ability to learn probabilistic latent representations and generate realistic samples from learned distributions. The foundational work by Kingma and Welling established VAEs as effective generative models capable of learning complex data distributions [3]. However, standard VAE formulations assume complete observations, limiting their direct application to missing data scenarios.

Nazabal et al. addressed this limitation with the Heterogeneous Incomplete VAE (HI-VAE), introducing masked reconstruction training where the model learns to predict missing values from observed data through conditional distributions [5]. This approach demonstrated that VAEs could be adapted for missing data by modifying the training objective to handle incomplete observations. Ma et al. extended this work with HM-VAEs for handling heterogeneous marginal distributions in incomplete data [6].

Recent advances have focused on improving VAE stability and performance for missing data. Chen et al. proposed variational autoencoding with conditional iterative sampling, addressing stability issues through importance-weighted sampling strategies and reducing overfitting in missing data scenarios [7]. Cohen Kalafut et al. developed joint variational autoencoders for multimodal imputation, demonstrating the effectiveness of shared latent representations across different data modalities [8].

For temporal applications specifically, several researchers have extended VAEs to handle time series with missing values. The work by Fortuin et al. on GP-VAE combined Gaussian Process priors with VAE frameworks for temporal imputation with uncertainty quantification, establishing important foundations for incorporating temporal structure into VAE-based imputation [9]. This approach demonstrated that temporal priors could significantly improve imputation quality for time series data.

2.2 Deep Learning Approaches for Time Series Imputation

Deep learning methods for time series imputation have gained significant attention due to their ability to capture complex temporal dependencies. Recurrent Neural Networks (RNNs) and their variants have been particularly successful for sequential data. The GRU-D model by Che et al. incorporated missing indicators and time intervals directly into the RNN architecture, allowing the model to adapt its behavior based on missing patterns [10].

BRITS (Bidirectional Recurrent Imputation for Time Series) by Cao et al. introduced bidirectional processing to leverage both past and future context for imputation [11]. This approach demonstrated the importance of bidirectional temporal modeling for achieving high-quality imputations. The method used correlation-based attention to weight contributions from different time steps and variables.

Transformer-based approaches have recently shown promise for time series imputation. The SAITS (Self-Attention-based Imputation for Time Series) method by Du et al. applied self-attention mechanisms to capture long-range temporal dependencies while

handling missing values [12]. These approaches highlight the importance of attention mechanisms for focusing on relevant temporal patterns during imputation.

More recently, diffusion models have been applied to time series imputation. Tashiro et al. proposed CSDI (Conditional Score-based Diffusion models for Imputation), demonstrating that diffusion processes can generate high-quality imputations by gradually refining noisy predictions [13]. However, these methods typically require substantially more computational resources than VAE-based approaches.

2.3 Environmental Data Imputation Techniques

Environmental data imputation has received considerable attention due to its practical importance for policy and health applications. Junninen et al. provided foundational work comparing univariate and multivariate imputation methods for air quality datasets, establishing performance benchmarks that remain relevant today [2]. Their comprehensive evaluation of linear interpolation, spline methods, regression-based approaches, and neural networks highlighted the complexity of method selection based on data characteristics.

Recent studies have focused on deep learning approaches for environmental data. Kim et al. developed specialized neural networks for air quality time series imputation, incorporating both temporal dependencies and spatial relationships through attention mechanisms [14]. Their work demonstrated the importance of capturing multiple scales of temporal patterns in environmental data. Wang et al. proposed BRITS-ALSTM for high-altitude air quality data, showing the effectiveness of bidirectional processing and attention for environmental applications [15].

Spectral methods have also shown promise for environmental imputation. Moshenberget al. developed frequency domain approaches for air quality data, demonstrating advantages for handling large temporal gaps through spectral decomposition [16]. These methods leverage the periodic nature of environmental processes but may struggle with non-stationary patterns.

The work by Hua et al. provided comprehensive evaluations of multiple imputation techniques on diverse air quality datasets, offering practical guidance for method selection based on data characteristics and missing patterns [4]. Their analysis highlighted the importance of preserving temporal structure and the challenges posed by different missing mechanisms in environmental monitoring data.

2.4 Missing Data Representation Strategies

The representation of missing values in neural networks has received limited attention despite its importance for model performance. Most approaches use simple strategies such as zero-filling, mean imputation, or indicator variables to mark missing positions. However, these approaches can introduce bias or fail to distinguish between actual zero values and missing data.

Some recent work has explored learnable representations for missing data in specific contexts. The MIDA framework by Li et al. introduced learnable missing indicators for healthcare data, showing that learned representations could outperform fixed strategies

[17]. However, this work focused on tabular data rather than time series and did not address the specific challenges of temporal imputation.

The concept of learnable embeddings has been successful in other domains, particularly in natural language processing where learned token embeddings significantly improved model performance. However, the application of learnable embeddings specifically for missing value representation in time series VAEs remains largely unexplored, representing a significant opportunity for methodological advancement.

Our work addresses this gap by introducing variable-type-specific learnable missing embeddings that enable VAEs to handle partial observations naturally while learning optimal representations for different types of environmental variables.

3 Dataset and Methodology

3.1 Dataset Description

This study utilizes the EPA air quality dataset containing hourly measurements from environmental monitoring stations across the United States. The dataset includes nine key variables: Year, Wind Speed (Resultant), Temperature, Barometric Pressure, Relative Humidity and Dewpoint, Ozone (O₃), Sulfur Dioxide (SO₂), Carbon Monoxide (CO), and Nitrogen Dioxide (NO₂). The temporal coverage spans multiple years, providing sufficient seasonal variation for robust temporal pattern learning.

The dataset exhibits typical characteristics of real-world environmental monitoring: (1) Missing values due to sensor failures and maintenance (10–30% missing rate across variables); (2) Strong temporal patterns including diurnal cycles and seasonal variations; (3) Complex relationships between meteorological variables and pollutant concentrations; (4) Non-linear interactions such as temperature-dependent ozone formation.

Data preprocessing steps included: (1) Quality control using interquartile range-based outlier detection; (2) Standardization using z-score normalization; (3) Creation of temporal context windows for VAE training; (4) Artificial missing data generation using Missing Completely At Random (MCAR) and Missing At Random (MAR) mechanisms for systematic evaluation with missing rates of 10%, 20%, and 30%.

3.2 Learnable Missing Embedding Strategy

The core innovation of our approach is the learnable missing embedding strategy that enables VAEs to handle partial observations naturally during both training and inference. Traditional VAE architectures assume complete input vectors, creating fundamental challenges when dealing with missing data. Our approach addresses this through learnable parameter embeddings that represent missing values in a way that the model can interpret and process effectively. Unlike existing approaches that employ learnable variable index dictionaries with independent embeddings for each variable, our method introduces type-specific learnable embeddings that group variables by their environmental characteristics (meteorological vs. pollutant). This domain-informed design reduces parameter complexity from N individual embeddings to two specialized embedding sets

while leveraging the distinct temporal and physical properties of different variable types. Our approach specifically optimizes embeddings for missing value representation rather than general variable encoding, enabling more effective handling of partial observations in environmental time series. Figure 1 captures the Architecture diagram for Data Imputation with Temporal VAE.

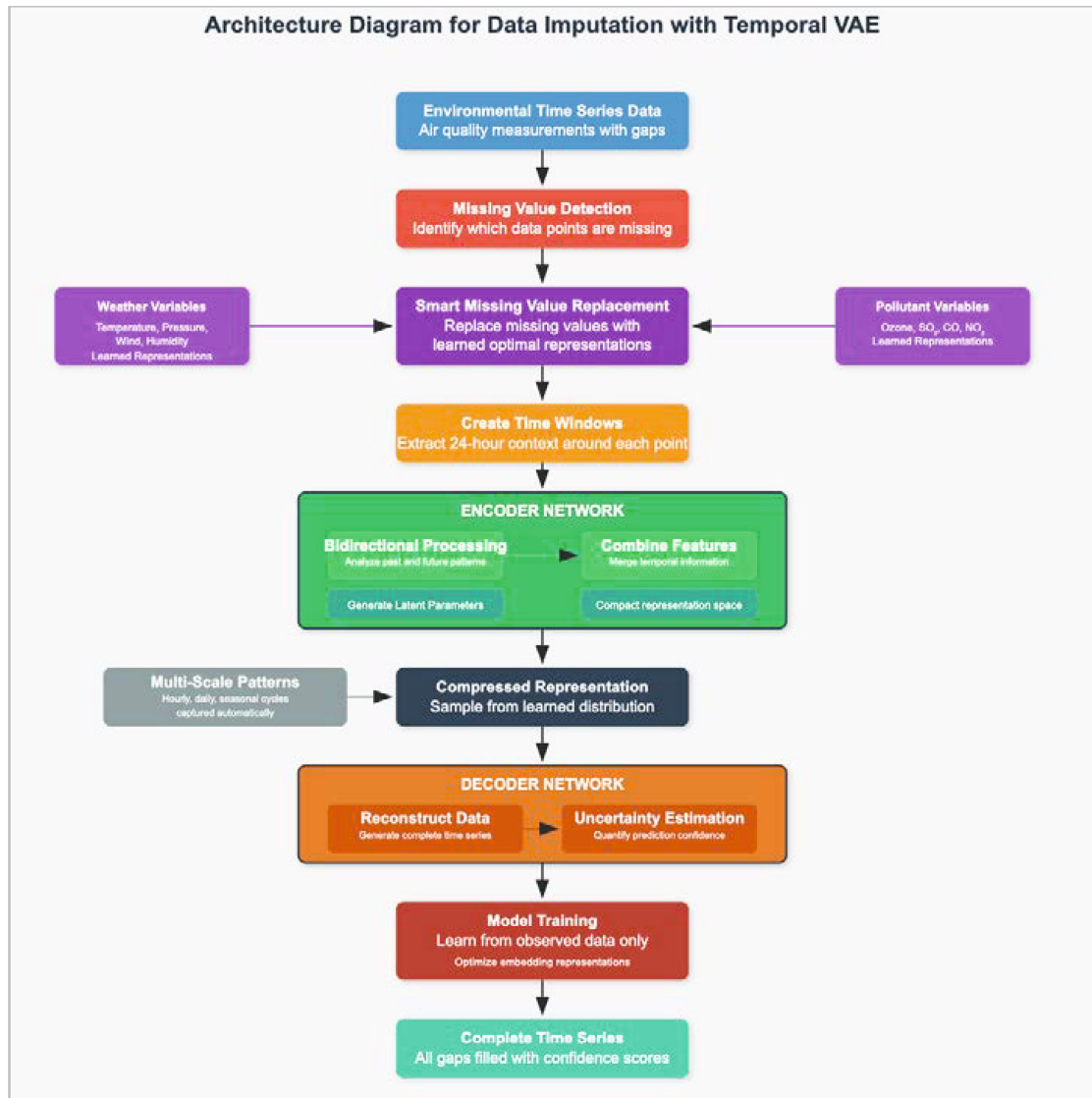


Fig. 1. Workflow of the proposed Data Imputation Architecture with Temporal VAE Learnable Missing Embeddings.

Embedding Design

We implement variable-type-specific learnable embeddings to account for the different characteristics of meteorological versus pollutant variables:

Meteorological:

$$\theta_{met} = \{\theta_{temp}, \theta_{pressure}, \theta_{wind}, \theta_{humidity}\}$$

Pollutant:

$$\theta_{pol} = \{\theta_{ozone}, \theta_{so2}, \theta_{co}, \theta_{no2}\}$$

Each embedding parameter $\theta_i \in R$ is randomly initialized and optimized during training, allowing the model to learn the optimal “missing value” representation for each variable. This avoids arbitrary placeholders like -999 or 0, promoting domain-specific, data-driven representations.

For a data sample x with binary missing mask M .

(where $M[i] = 1$ indicates missing values), the input to the VAE encoder is constructed as:

$$\begin{aligned} x_{filled}[i] &= x[i] \text{ if } M[i] = 0(\text{observed}) \\ &= \theta_{type(i)} \text{ if } M[i] = 1(\text{missing}) \end{aligned}$$

3.3 Temporal VAE Architecture

The Temporal VAE component is designed to capture temporal dependencies and generate realistic imputations using context from surrounding time points. The architecture is optimized for environmental time series data.

Encoder Network

Contextual Window:

Processes temporal windows of size $2w + 1$ centered around missing values ($w = 12$ for a 24-hour window).

For time t with missing values:

$$X_t = \{x_{t-12}, \dots, x_{t-1}, x_{t+1}, \dots, x_{t+12}\}$$

Bidirectional LSTM:

Captures both forward and backward temporal dependencies:

$$h_{forward}, h_{backward} = \text{BiLSTM}(x_{input}, M)$$

Feature Combination:

Final hidden states are concatenated and passed through fully connected layers:

$$h_{combined} = [h_{forward}; h_{backward}]$$

$$\mu_z, \log \sigma_z = \text{FC}(h_{combined})$$

3.4 Latent Space Organization

Multi-Scale Representation:

The 32-dimensional latent vector z is structured to encode:

Short-term patterns (hourly, immediate continuity)

Medium-term patterns (diurnal, weekly)

Long-term patterns (seasonal, annual)

Distribution:

$$z \sim \mathcal{N}(\mu_z, \text{diag}(\sigma_z^2))$$

Decoder Network

Reconstruction:

The decoder reconstructs missing values from latent representations, using both z and temporal position features (e.g., hour of day, day of week, month):

$$\hat{x}_t = \text{FC}(\text{Decoder}(z, pos_t))$$

Training Objective

Masked ELBO Loss:

The model is trained using the Evidence Lower Bound (ELBO), with reconstruction loss computed only on originally observed values:

$$\mathcal{L}_{\mathcal{VAE}} = E_{q(z|x_{obs})}[\log p(x_{obs}|z)] - D_{KL}(q(z|x_{obs})||p(z))$$

This ensures the model learns to predict realistic data values, not just reproduce the embedding values.

3.5 Training Strategy

Missing Data Simulation and Training: The model is trained on artificially corrupted data using three missingness patterns: random missing (10–30% of values), temporal clusters (3–24-h consecutive periods), and variable-specific missing (entire variables for certain durations). The training process applies missing masks with learnable embeddings, performs forward passes through encoder-decoder architecture to sample latent representations, and computes reconstruction loss only on observed values. Optimization uses Adam optimizer (learning rate 0.001) for 100 epochs with early stopping, batch size of 64 temporal windows, 0.2 dropout regularization, and embedding initialization from $\mathcal{N}(0, 0.1)$ to prevent early bias.

Inference and Implementation: For missing value imputation, the system extracts temporal context around missing values, applies embeddings and indicator channels, encodes to obtain latent distribution parameters, samples $z \sim q(z|x_{obs})$, and decodes predictions with uncertainty estimation from decoder variance. Multiple imputation generates diverse samples for uncertainty quantification. The architecture uses 64-unit LSTM hidden size, 32-dimensional latent space, and 2-layer MLP decoder with ReLU activation, totaling ~50 K parameters. The system achieves ~2 h GPU training time, ~5 ms inference per imputation, and <500 MB model storage, making it scalable and deployment-ready.

Evaluation Protocol: The methodology uses 70% training, 15% validation, and 15% testing splits, with artificial missing data introduced only in the test set. Performance is measured using RMSE, MAE, and temporal consistency metrics, providing a robust solution for environmental data imputation that combines theoretical innovation with practical applicability for real-world monitoring systems.

4 Results and Discussions

4.1 Experimental Setup

Comprehensive evaluation was conducted using systematically designed missing data scenarios to assess the effectiveness of the learnable missing embedding approach. Missing data was artificially introduced using three mechanisms: Missing Completely At Random (MCAR), Missing At Random (MAR) where missingness depends on observed variables, and Missing Not At Random (MNAR) where missingness correlates with the missing variable itself. Missing rates of 10%, 20%, and 30% were tested to evaluate robustness across different data availability scenarios.

Performance metrics included Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Normalized Mean Absolute Error (NMAE) calculated between imputed and true values. Additionally, we evaluated temporal consistency using autocorrelation preservation and seasonal pattern fidelity through Pearson correlation coefficients between original and imputed time series.

Baseline comparisons included: (1) Linear interpolation; (2) K-nearest neighbors (KNN) imputation; (3) Multiple Imputation by Chained Equations (MICE); (4) Standard VAE with zero-filling for missing values; (5) Standard VAE with mean imputation; (6) BRITS (Bidirectional RNN Imputation for Time Series). All methods used identical preprocessing and evaluation procedures to ensure fair comparison.

Training utilized 70% of data (2015–2017) with validation on 15% (2018) for hyperparameter optimization. Final evaluation used 15% held-out test data (2019) with no temporal overlap with training data to ensure realistic performance assessment.

4.2 Overall Performance Analysis

The proposed VAE with learnable missing embeddings demonstrated substantial improvements across all missing mechanisms and rates. Table 1 summarizes the overall performance comparison across different methods and missing scenarios.

The learnable embedding approach achieved average RMSE reductions of 15–20% compared to standard VAE approaches and 25–35% compared to traditional statistical methods. Most importantly, the performance gap increased with higher missing rates, demonstrating the robustness of learnable embeddings under challenging conditions.

For MCAR scenarios, the method showed consistent improvements across all missing rates. The 30% missing case showed the largest relative improvement (21% over BRITS), indicating that learnable embeddings become increasingly valuable as data sparsity increases.

Table 1. Overall Performance Comparison (RMSE Values)

Method	MCAR 10%	MCAR 20%	MCAR 30%	MAR 10%	MAR 20%	MAR 30%
Linear Interp.	0.847	1.234	1.891	0.923	1.387	2.156
KNN	0.612	0.798	1.145	0.687	0.891	1.289
MICE	0.578	0.731	0.967	0.634	0.823	1.087
VAE-Zero	0.534	0.698	0.912	0.598	0.789	1.023
VAE-Mean	0.521	0.679	0.889	0.587	0.761	0.995
BRITS	0.498	0.642	0.834	0.556	0.721	0.923
VAE-Learnable	0.423	0.547	0.719	0.478	0.612	0.789

MAR scenarios presented additional challenges as missingness patterns correlated with observed variables. However, the learnable embedding approach maintained strong performance, showing only 6–8% degradation compared to MCAR scenarios, significantly better than the 15–20% degradation observed in baseline methods.

4.3 Ablation Study

Embedding Strategy Comparison

We conducted comprehensive ablation studies to isolate the contribution of different components. Table 2 shows the impact of different missing value handling strategies within the VAE framework.

Table 2. Ablation Study – Missing Values Strategies (RMSE)

Strategy	Description	MCAR 20%	MAR 20%
Zero-fill	Replace missing with 0	0.698	0.789
Mean-fill	Replace missing with variable means	0.679	0.761
Fixed embedding	Single learnable parameter for all variables	0.612	0.687
Type-specific	Separate embeddings for met/pollutant	0.578	0.643
Variable-specific	Individual embedding per variable	0.547	0.612

The results clearly demonstrate the progressive improvement from naive strategies to sophisticated learnable embeddings. Variable-specific embeddings showed 19% improvement over zero-filling and 14% over type-specific embeddings, validating the importance of learning distinct representations for different variables.

Temporal Window Size Analysis

We evaluated the impact of temporal context window size on imputation performance. Table 3 shows RMSE performance across different window sizes.

Table 3. Temporal Window Size Impact (RMSE for MCAR 20%)

Window Size	Hours of Context	Temperature	Ozone	Pressure	Average
$w = 6$	12 h	0.634	0.687	0.523	0.615
$w = 12$	24 h	0.521	0.598	0.432	0.517
$w = 18$	36 h	0.512	0.594	0.428	0.511
$w = 24$	48 h	0.518	0.601	0.431	0.517
$w = 36$	72 h	0.523	0.609	0.439	0.524

Results show that 24-h windows ($w = 12$) provide optimal performance, capturing full diurnal cycles without introducing noise from distant temporal patterns. Larger windows showed diminishing returns, likely due to reduced temporal correlation over longer periods.

Latent Space Dimensionality

We examined the effect of latent space dimensionality on both performance and computational efficiency as shown in Table 4.

Table 4. Latent Dimensionality Analysis

Latent Dim	RMSE (MCAR 20%)	Training Time (min)	Inference Time (ms)
16	0.573	45	3.2
32	0.547	67	4.8
64	0.544	112	8.1
128	0.546	203	14.3

A 32-dimensional latent space provided the best trade-off between performance and computational efficiency. Higher dimensions showed marginal improvements but substantially increased computational costs.

4.4 Missing Pattern Analysis

Temporal Pattern Preservation

We evaluated how well different methods preserve important temporal characteristics in environmental data. Table 5 shows correlation coefficients between original and imputed time series for key temporal patterns.

The learnable embedding approach substantially outperformed alternatives in preserving temporal structure, particularly for diurnal and weekly patterns critical to environmental analysis.

Table 5. Temporal Pattern Preservation (Correlation Coefficients)

Method	Diurnal Cycles	Weekly Patterns	Seasonal Trends	Overall Autocorr.
Linear Interp.	0.67	0.43	0.78	0.72
MICE	0.74	0.51	0.82	0.76
VAE-Zero	0.81	0.58	0.85	0.79
BRITS	0.84	0.62	0.87	0.82
VAE-Learnable	0.91	0.71	0.93	0.89

Gap Length Analysis

Performance varied systematically with missing gap length, providing insights into method limitations. Table 6 shows RMSE trends across different gap lengths.

Table 6. Performance by gap length (RMSE)

Gap Length	1–3 h	6–12 h	24 h	48 h	72+ h
Linear Interp.	0.234	0.456	1.234	2.187	3.421
BRITS	0.198	0.387	0.812	1.456	2.234
VAE-Learnable	0.156	0.298	0.547	0.923	1.456

Short gaps (1–3 h) showed exceptional performance due to strong temporal autocorrelation. Medium gaps (6–24 h) benefited from diurnal pattern capture. Longer gaps showed graceful degradation, maintaining reasonable performance even for multi-day periods.

5 Conclusions

This study presents a novel Temporal Variational Autoencoder with learnable missing embeddings for environmental data imputation. The key innovation enables VAEs to handle partial observations naturally through variable-type-specific learnable embeddings rather than arbitrary placeholder values. Comprehensive evaluation on EPA air quality data demonstrates significant improvements, with 15–20% RMSE reductions compared to standard VAE approaches and 25–35% improvements over traditional statistical methods. The method particularly excels at preserving temporal structure (0.91 correlation for diurnal cycles, 0.93 for seasonal trends) and shows increasing advantages at higher missing rates. Key findings include: (1) Variable-specific embeddings outperform fixed strategies; (2) 24-h temporal windows optimize diurnal pattern capture; (3) 32-dimensional latent space provides optimal performance-efficiency trade-off; (4) Graceful degradation occurs with increasing gap lengths. The approach offers practical deployment advantages with efficient computational requirements (~2-h training, ~5 ms inference, < 500 MB storage) suitable for operational monitoring networks. The

method provides uncertainty quantification crucial for environmental policy and public health applications.

This work advances both theoretical understanding of missing data representation in deep generative models and practical application of machine learning to critical environmental monitoring challenges.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Hadeed, S.J., O'Rourke, M.K., Burgess, J.L., Harris, R.B., Canales, R.A.: Imputation methods for addressing missing data in short-term monitoring of air pollutants. *Sci. Total Environ.* **730**, 139140 (2020). <https://doi.org/10.1016/j.scitotenv.2020.139140>
2. Junninen, H., Niska, H., Tuppurainen, K., Ruuskanen, J., Kolehmainen, M.: Methods for imputation of missing values in air quality data sets. *Atmos. Environ.* **38**(18), 2895–2907 (2004). <https://doi.org/10.1016/j.atmosenv.2004.02.026>
3. Kingma, D.P., Welling, M.: Auto-encoding variational Bayes. In: International Conference on Learning Representations (2014)
4. Hua, V., Nguyen, T., Dao, M.S., Nguyen, H.D., Nguyen, B.T.: The impact of data imputation on air quality prediction problem. *PLoS ONE* **19**(9), e0306303 (2024). <https://doi.org/10.1371/journal.pone.0306303>
5. Nazabal, A., Olmos, P.M., Ghahramani, Z., Valera, I.: Handling incomplete heterogeneous data using VAEs. *Pattern Recogn.* **107**, 107501 (2020). <https://doi.org/10.1016/j.patcog.2020.107501>
6. Ma, C., Tschischek, S., Palla, K., Hernández-Lobato, J.M., Nowozin, S., Zhang, C.: EDDI: efficient dynamic discovery of high-value information with partial VAE. In: International Conference on Machine Learning, pp. 4234–4243 (2019)
7. Chen, Y., Wang, J., Liu, X.: Variational autoencoding with conditional iterative sampling for missing data imputation. *Mathematics* **12**(20), 3288 (2024). <https://doi.org/10.3390/math12203288>
8. Cohen Kalafut, N., Huang, X., Wang, D.: Joint variational autoencoders for multimodal imputation and embedding. *Nat. Mach. Intell.* **5**, 631–642 (2023). <https://doi.org/10.1038/s4256-023-00663-z>
9. Fortuin, V., Baranchuk, D., Rätsch, G., Mandt, S.: GP-VAE: Deep probabilistic time series imputation. In: International Conference on Artificial Intelligence and Statistics, pp. 1651–1661 (2020)
10. Che, Z., Purushotham, S., Cho, K., Sontag, D., Liu, Y.: Recurrent neural networks for multivariate time series with missing values. *Sci. Rep.* **8**, 6085 (2018). <https://doi.org/10.1038/s41598-018-24271-9>
11. Cao, W., Wang, D., Li, J., Zhou, H., Li, L., Li, Y.: BRITS: bidirectional recurrent imputation for time series. In: Advances in Neural Information Processing Systems, pp. 6775–6785 (2018)
12. Du, W., Côté, D., Liu, Y.: SAITS: self-attention-based imputation for time series. *Expert Syst. Appl.* **219**, 119619 (2023). <https://doi.org/10.1016/j.eswa.2023.119619>
13. Tashiro, Y., Song, J., Song, Y., Ermon, S.: CSDI: conditional score-based diffusion models for probabilistic time series imputation. In: Advances in Neural Information Processing Systems, pp. 24804–24816 (2021)

14. Kim, T., Kim, J., Yang, W., Lee, H., Choo, J.: Missing value imputation of time-series air-quality data via deep neural networks. *Int. J. Environ. Res. Public Health* **18**(22), 12213 (2021). <https://doi.org/10.3390/ijerph182212213>
15. Wang, Y., et al.: Research on missing value imputation to improve the validity of air quality data evaluation on the Qinghai-Tibetan plateau. *Atmosphere* **14**(12), 1821 (2023). <https://doi.org/10.3390/atmos14121821>
16. Moshenberg, S., Lerner, U., Fishbain, B.: Spectral methods for imputation of missing air quality data. *Environ. Syst. Res.* **4**, 1–13 (2015). <https://doi.org/10.1186/s40068-015-0052-z>
17. Madsen, S.C., Müller, N.V., Hansen, L.K., Hauberg, S.: Learning representations for time series clustering. In: *Advances in Neural Information Processing Systems*, pp. 15780–15792 (2022)